

Positive Real funcⁿ

(0)

$T(s) = \frac{N(s)}{D(s)} \rightarrow$ P.R. if follⁿ condⁿ are satisfied

- (1) $T(s)$ is real for s real.
- (2) $D(s)$ is Hurwitz polynomial.
- (3) $T(s)$ may have poles on the $j\omega$ axis.

(4) Real part of $T(s)$ is greater than or equal to 0 for real part of $s \geq 0$.

$$\operatorname{Re} [T(s)] \geq 0 \quad \operatorname{Re} s \geq 0$$

$$T(s) = \frac{N(s)}{D(s)} = \frac{M_1(s) + N_1(s)}{M_2(s) + N_2(s)}$$

$$T(s) = \frac{M_1 + N_1}{M_2 + N_2} \cdot \frac{M_2 - N_2}{M_2 - N_2}$$

$$= \frac{M_1 M_2 - N_1 N_2}{M_2^2 - N_2^2} \cdot \frac{N_1 M_2 - M_1 N_2}{M_2^2 - N_2^2}$$

$$\operatorname{Ev} [T(s)] = \frac{M_1 M_2 - N_1 N_2}{M_2^2 - N_2^2}$$

$$\operatorname{odd} [T(s)] = \frac{N_1 M_2 - M_1 N_2}{M_2^2 - N_2^2}$$

$$s = j\omega.$$

even part $\rightarrow$ real

odd part $\rightarrow$ imag.

$$\operatorname{Re} [T(j\omega)] = \operatorname{Ev} [T(s)]|_{s=j\omega}.$$

$$j \operatorname{Im} [T(j\omega)] = \operatorname{odd} [T(s)]|_{s=j\omega}.$$

So To prove this condn.

$$\operatorname{Re} [T(j\omega)] \geq 0 \text{ for all } \omega.$$

$$\text{or } M_1(j\omega) \cdot M_2(j\omega) - N_1(j\omega) N_2(j\omega) \geq 0.$$

Since Denominator is always +ve.

Properties:

$$(1) \quad T(s) \rightarrow \text{p.s.} \text{ then } \frac{1}{T(s)} \rightarrow \text{p.s.}$$

$$\text{If } Z(s) \rightarrow \text{p.s.}$$

then $Y(s) \rightarrow$ also p.s.

$$(2) \quad \text{Sum of p.s. f}^n \rightarrow \text{p.s.}$$

(3) Poles & zeros of a p.s.f can't be in the right half of s-plane.

in addition only simple poles with $\text{Re}(s) < 0$ real the residues can exist on $j\omega$ -axis.

(4) poles & zeros of a p.r.f are real or occur in conjugate pair.

(5) highest power of $N(s)$ & $D(s)$ may differ atmost by unity.
 $\Rightarrow$ prohibits multiple poles or zeros at ∞ .

(6) lowest power of $D(s)$ & $N(s)$ may differ by atmost unity.
 $\Rightarrow$ prohibits multiple poles or z at $s=0$.

Necessary & sufficient condⁿ for $T(s)$ to be P.r.

(1) $D(s)$ must be Hurwitz.

(2) This condⁿ is checked only when the poles of $T(s)$ are on the $j\omega$ axis. otherwise not.

* is tested by making partial fr. expansion of $T(s)$ & checking whether the residues of the poles on $j\omega$ -axis are +ve & real.

Cardt $\rightarrow \operatorname{Re}[T(j\omega)] \geq 0$ for all ω .

$$M_1(j\omega) M_2(j\omega) - N_1(j\omega) N_2(j\omega) \geq 0.$$

Q show that $f(s) = \frac{(s+2)(s+4)}{(s+1)(s+3)}$ is p.o.s.
$$= \frac{s^2 + 6s + 8}{s^2 + 4s + 3}$$

Sol (1) $D(s) \rightarrow$ Hurwitz.

(2) $\rightarrow$ doesn't exist.

$$(3) \quad M_1 = s^2 + 8 \quad N_1 = 6s$$

$$M_2 = s^2 + 3 \quad N_2 = 4s.$$

$$M_1 M_2 - N_1 N_2 \geq 0$$

$$(s^2 + 8)(s^2 + 3) - (6s)(4s) \geq 0$$

$$s^4 - 13s^2 + 24 \geq 0$$

$$A(\omega) = \omega^4 + 13\omega^2 + 24 \geq 0 \text{ for all } \omega.$$

$$\frac{-3s}{s^2+1} + \frac{5}{s} \Big|_{s=j\omega} = \frac{-3j}{j^2+1} + \frac{5}{j} = \text{---}$$

$$\underline{8} \quad f(s) = \frac{s^5 + s^2 + 3s + 5}{s^2 + 6s + 8} \quad (c)$$

Ex 1
Guidtⁿ : $M(s) = s^2 + 8$ $N(s) = 6(s)$.
 $D(s) \rightarrow$ Hurwitz.

(2) $\rightarrow$ no poles on $j\omega$.

(3) $M_1 = s^2 + 5$ $M_2 = s^2 + 8$.

$N_1 = s^3 + 3s$ $N_2 = 6s$.

$A(\omega^2) = M_1 M_2 - N_1 N_2 \geq 0$.

$(s^2 + 5)(s^2 + 8) - (s^3 + 3s)(6s) \geq 0$.

$-5s^4 - 5s^2 + 40 \geq 0$

$-5\omega^4 + 5\omega^2 + 40 \geq 0 \quad (s = j\omega)$.

Since all coeff. $\neq$ not +ve.

So we use Sturm's test.
 Put $\omega^2 \Rightarrow x$

$A_0(x) = -5x^2 + 5x + 40$.

$A_1(x) = \frac{dA_0(x)}{dx} = -10x + 5$

$\frac{A_2(x)}{A_1(x)} = \frac{(K_1x - K_0) - A_2(x)}{A_1(x)} =$

$$\begin{array}{r} \cancel{A_0} \quad -10x+5 \quad -5x^2+5x+40 \quad \left(\frac{x}{2} - \frac{1}{4} \right) \\ -5x^2 + \frac{5x}{2} \\ \hline \end{array}$$

$$\begin{array}{r} \frac{5x+40}{2} \\ \frac{5x}{2} + \frac{15}{4} \\ \hline 165 \\ 4 \end{array}$$

$$\frac{A_0(x)}{A_1(x)} = \left(\frac{x}{2} - \frac{1}{4} \right) + \frac{165/4}{-10x+5}$$

$$A_2 x = -165/4$$

	A_0	A_1	A_2	No. of sign change.
$x=0$	+	+	-	$S_0=1$
$x=\infty$	-	-	-	$S_\infty=0$

$$S_\infty \sim S_0 = 1$$

$\therefore A(\omega^2) \not\equiv 0$ for all ω .

hence given funcⁿ is not a p.r.f.

Q Determine whether the funcⁿ is p.r. or not (a)

$$Z(s) = \frac{2s^2 + 5}{s(s^2 + 1)}$$

(1) $D(s) = s^3 + s$ then $D'(s) = 3s^2 + 1$

$D(s) \rightarrow$ Hurwitz.

(2) $Z(s) \rightarrow$ poles $\Rightarrow s = \pm j1$

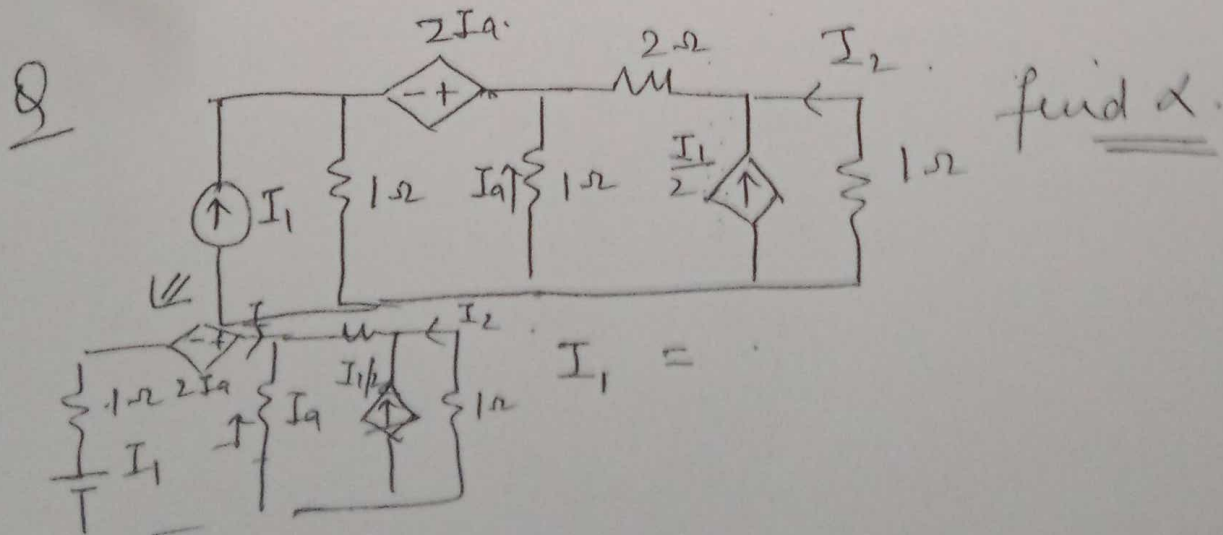
$$\begin{aligned} s(s^2 + 1) \\ s = 0 \\ s^2 + 1 = 0 \\ s^2 = -1 \\ s = \pm j1 \end{aligned}$$

$$Z(s) = \frac{-3s}{s^2 + 1} + \frac{5}{s}$$

$$Zs \Big|_{s^2 = -1} = \frac{2s^2 + 5}{s(s^2 + 1)} \Big|_{s^2 = -1} = \frac{-2 + 5}{-1} = -3$$

Since residue of the poles at $s = \pm j1$ is -ve.

So it's not p.r.f.



Foster & Cauer forms :

①
⑧

Synthesis of 1 port N/w with 2 kinds of element

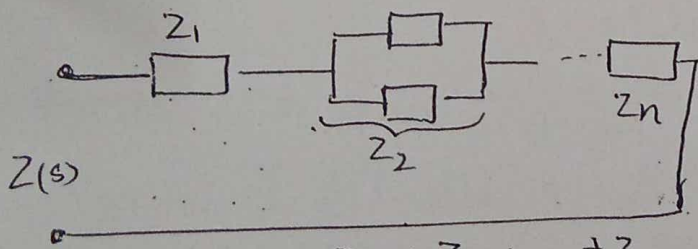
Since we have R, L, C so.

N/w can be synthesized either as

- 1) L-C
- 2) R-L
- 3) R-C

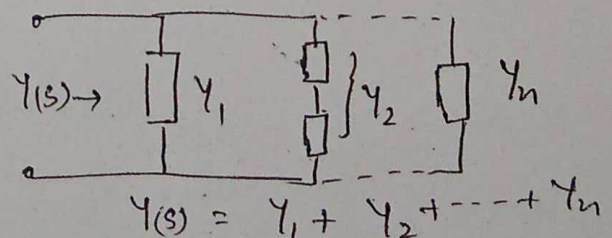
methods

- 1) Foster - I or foster-series form
 - 2) Foster - II or " parallel "
 - 3) Cauer - I form
 - 4) Cauer - II form
- { obtained by partial fraction expansion of $Z(s)$ & $Y(s)$ resp }
- { forms by Contd. fraction expansion by arranging both numerator & denominator in descending & ascending order resp }



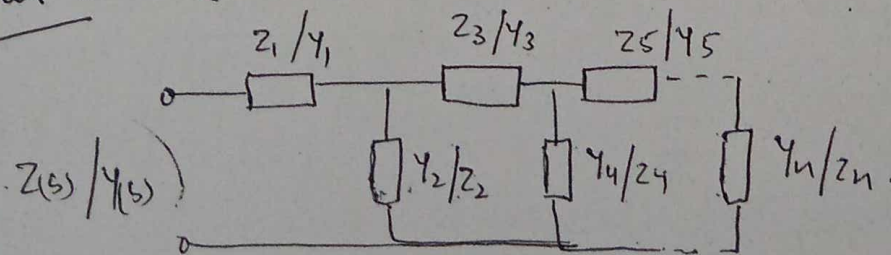
$$Z(s) = Z_1 + Z_2 + \dots + Z_n$$

Foster-I (a)



$$Y(s) = Y_1 + Y_2 + \dots + Y_n$$

(b) Foster-II



Cauer form

L-C Impedance funcⁿ: $Z(s) = \frac{M_1(s) + N_1(s)}{M_2(s) + N_2(s)}$

Properties

- ① $Z_{LC}(s)$ or $Y_{LC}(s)$ are the ratio of even to odd or odd to even polynomials.
- ② Since $M_i(s)$ & $N_j(s)$ are Hurwitz, they have only imaginary roots, & it follows that P & Z of Z_{LC} & Y_{LC} are on $j\omega$ axis.
- ③ P & Z interlace on the $j\omega$ axis. (alternate).
- ④ Highest Powers of Num & Den. must differ by unity.
- ⑤ Lowest Power of Num & Den. must also differ by unity.
- ⑥ There must be either a zero or a pole at the origin & ∞ .

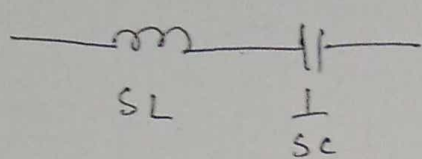
Q Find if the following funcⁿ are L-C impedance funcⁿ or not. If it is plot its P & Z.

① ~~$\frac{s^5 + 3s^3 + 9s}{7s^4 + 6s^2}$~~
 $\frac{Ks(s^2+4)}{(s^2+1)(s^2+3)}$
 (not L-C)

② $\frac{2(s^2+1)(s^2+9)}{s(s^2+4)}$ ✓
 (is L-C)

(3)

L-C in series



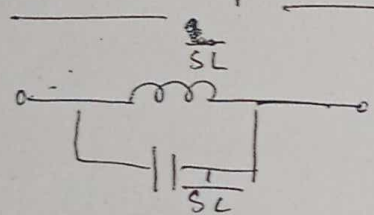
$$Z(s) = sL + \frac{1}{sc}$$

$$= \frac{s^2 Lc + 1}{sc}$$

$$= \frac{s^2 + \frac{1}{Lc}}{\frac{s}{L}}$$

$$Y(s) = \frac{\frac{1}{L}s}{s^2 + \frac{1}{Lc}}$$

L-C in parallel



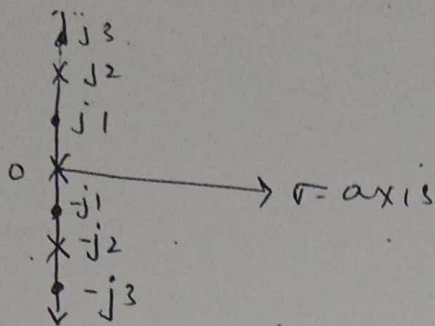
$$Z(s) = \frac{1}{\frac{1}{sL} + sc}$$

$$= \frac{s^2 + \frac{1}{Lc}}{\frac{s}{c}}$$

$$Z(s) = \frac{1}{Y(s)} = \frac{-\frac{1}{c} \cdot s}{s^2 + \frac{1}{Lc}}$$

Ques An imp. funcⁿ has the pole zero pattern as shown. If $Z(2) = -\frac{130}{16}$ Synthesize the imp. in (a) Foster-I & II forms (b) Cauer-I & II forms.

Solⁿ



$$Z(s) = \frac{K(s^2+1)(s^2+9)}{s(s^2+4)}$$

$$Z(-2) = \frac{-130}{16} = \frac{K \cdot 5 \cdot 13}{-2 \cdot 8}$$

$$\Rightarrow K = 2$$

$$Z(s) = 2 \frac{(s^2+1)(s^2+9)}{s(s^2+4)}$$

a) Foster - I form

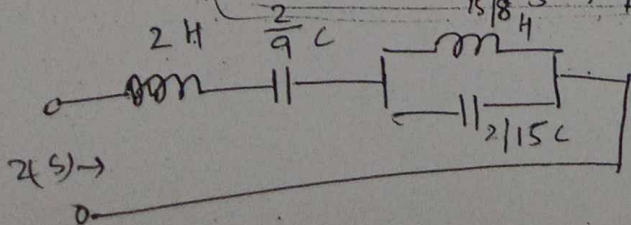
$$\begin{array}{l} s^3 + 4s \\ 2s^4 + 20s^2 + 18 \\ \hline 2s^4 + 8s^2 \\ \hline 12s^2 + 18 \end{array} \quad \left[\begin{array}{l} 2s \\ 16/3 \\ 3/16 \end{array} \right] \rightarrow Y(s)$$

$$Z(s) = 2s + \frac{12s^2 + 18}{s(s^2+4)}$$

$$\frac{12s^2 + 18}{s(s^2+4)} = \frac{A}{s} + \frac{Bs+C}{s^2+4}$$

$$A = 9/2 \quad C = 0 \quad B = \frac{15}{2}$$

$$Z(s) = 2s + \frac{9}{2s} + \frac{15 \cdot s}{2(s^2+4)}$$



Foster - II

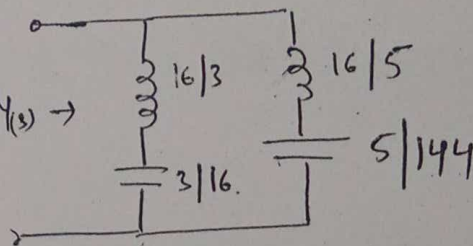
$$Y(s) = \frac{s(s^2+4)}{2(s^2+1)(s^2+9)}$$

$$= \frac{1}{2} \left[\frac{As+B}{s^2+1} + \frac{Cs+D}{s^2+9} \right]$$

$$A = \frac{3}{8} \quad B = D = 0 \quad C = \frac{5}{8}$$

$$Y(s) = \frac{1}{2} \left[\frac{\frac{3}{8}s}{s^2+1} + \frac{\frac{5}{8}s}{s^2+9} \right]$$

$$Y(s) = \frac{\frac{3}{16}s}{s^2+1} + \frac{\frac{5}{16}s}{s^2+9}$$



$$\frac{1}{C} = L$$

b Cauer - I form :

(3)

$$Z(s) = \frac{2s^4 + 20s^2 + 18}{s^3 + 4s}$$

$$s^3 + 4s \Big) 2s^4 + 20s^2 + 18 \quad (2s \leftarrow Z_1)$$

$$2s^4 + 8s^2$$

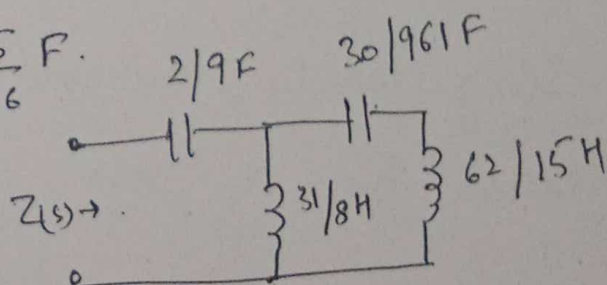
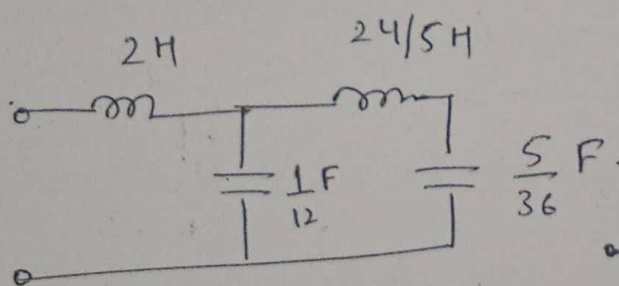
$$\hline 12s^2 + 18 \quad (12s^2 \leftarrow Z_2)$$

$$s^3 + \frac{3}{2}s$$

$$\hline \frac{5}{2}s \quad (12s^2 + 18 \leftarrow Z_3)$$

$$\hline 18 \quad (\frac{5}{2}s \leftarrow Z_4)$$

$$\frac{5}{2}$$



Cauer - II form

$$Z(s) = \frac{18 + 20s^2 + 2s^4}{s^3 + 4s}$$

$$4s + s^3 \Big) 18 + 20s^2 + 2s^4 \quad \left(\frac{18}{4s} = \frac{9}{2s} \rightarrow Z_1 \right)$$

$$18 + \frac{9}{2}s^2$$

$$\hline \frac{31}{2}s^2 + 2s^4 \quad (4s + s^3 \leftarrow Z_2)$$

$$4s + \frac{16}{31}s$$

$$\hline \frac{15}{31}s^3 \quad (\frac{31}{2}s^2 + 2s^4 \leftarrow Z_3)$$

$$\hline \frac{15}{31}s^3 \quad (\frac{1}{2} \cdot \frac{15}{31} = \frac{15}{62} \rightarrow Z_4)$$

Foster - II

$$Y(s) = \frac{s(s+2)(s+5)}{(s+1)(s+4)}$$

$$\frac{Y(s)}{s} = \frac{(s+2)(s+5)}{(s+1)(s+4)} = \frac{s^2 + 7s + 10}{s^2 + 5s + 4}$$

$$= 1 + \frac{2s+6}{(s+1)(s+4)}$$

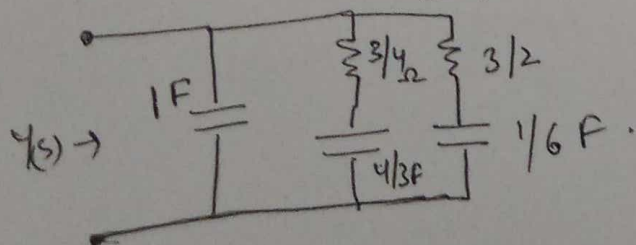
$$\frac{2s+6}{(s+1)(s+4)} = \frac{A}{s+1} + \frac{B}{s+4}$$

$$A = \left. \frac{2s+6}{s+4} \right|_{s=-1} = \frac{-2+6}{-1+4} = \frac{4}{3}$$

$$B = \left. \frac{2s+6}{s+1} \right|_{s=-4} = \frac{-8+6}{-4+1} = \frac{-2}{-3} = \frac{2}{3}$$

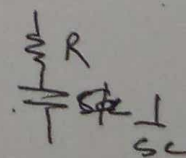
$$\frac{Y(s)}{s} = 1 + \frac{\frac{4}{3}}{s+1} + \frac{\frac{2}{3}}{s+4}$$

$$Y(s) = s + \frac{\frac{4}{3}s}{s+1} + \frac{\frac{2}{3}s}{s+4}$$



$$\frac{1}{RC} = 4$$

$$C = \frac{1}{R \times 4} = \frac{2}{3 \times 4}$$



$$\frac{sc}{Rsc+1}$$

$$Z = R + \frac{1}{sc} \\ = \frac{Rsc+1}{sc}$$

(b) Cauer - I form :

$$Z(s) = \frac{(s+1)(s+4)}{s(s+2)(s+5)} = \frac{s^2 + 5s + 4}{s^3 + 7s^2 + 10s}$$

$$Y(s) = \frac{s^3 + 7s^2 + 10s}{s^2 + 5s + 4}$$

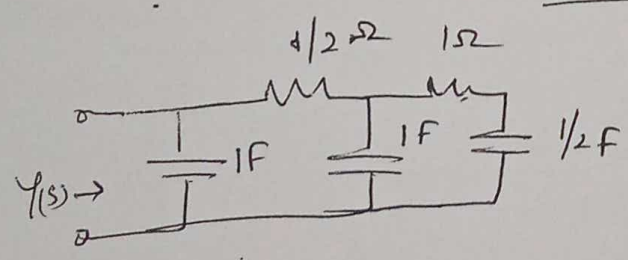
$$\begin{array}{r} s^2 + 5s + 4 \overline{) s^3 + 7s^2 + 10s} \quad (s \leftarrow Y_1) \\ \underline{s^3 + 5s^2 + 4s} \\ 2s^2 + 6s \end{array}$$

$$\begin{array}{r} s^2 + 5s + 4 \overline{) s^2 + 5s + 4} \quad (\frac{1}{2} \leftarrow Z_2) \\ \underline{s^2 + 3s} \\ 2s + 4 \end{array}$$

$$\begin{array}{r} 2s + 4 \overline{) 2s^2 + 6s} \quad (s \leftarrow Z_3) \\ \underline{2s^2 + 4s} \\ 2s + 4 \end{array}$$

$$\begin{array}{r} 2s \overline{) 2s + 4} \quad (1 \leftarrow Z_4) \\ \underline{2s} \\ 4 \end{array}$$

$$\begin{array}{r} 4 \overline{) 2s} \quad (\frac{1}{2} s \leftarrow Y_5) \\ \underline{2s} \\ 0 \end{array}$$

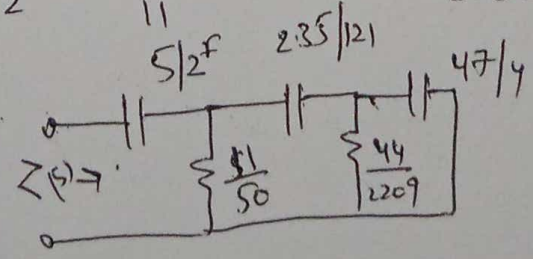


Cauer - II form :

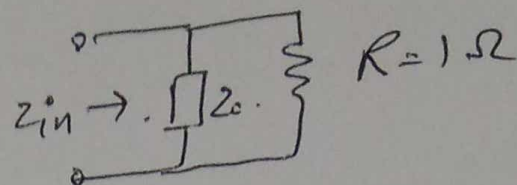
$$Z(s) = \frac{s^2 + 5s + 4}{s^3 + 7s^2 + 10s} = \frac{4 + 5s + s^2}{10s + 7s^2 + s^3}$$

$$Z_1 \rightarrow \frac{2}{5s} \quad Y_2 = \frac{50}{11} \quad Z_3 \rightarrow \frac{121}{235s} \quad Y_4 \rightarrow \frac{2209}{44}$$

$$Z_5 \rightarrow \frac{4}{47s}$$



$$Z_{in} = \frac{2s^2 + 2}{s^3 + 2s^2 + 2s + 2}$$



if $Z_0 =$ LC Imp. funⁿ find Z_0 .

(b) Synthesize Z_0 in a Foster-I form.

Soln

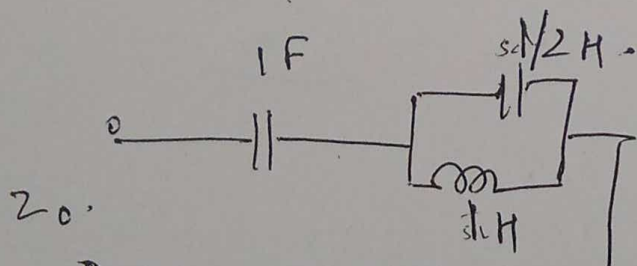
$$Z_{in} = \frac{Z_0 R}{Z_0 + R} = \frac{R}{1 + \frac{R}{Z_0}}$$

$$Z_{in} = \frac{2s^2 + 2}{s^3 + 2s^2 + 2s + 2}$$

$$1 + \frac{s^3 + 2s}{2s^2 + 2}$$

$$Z_0 = \frac{2s^2 + 2}{s^3 + 2s}$$

$$Z_0 = \frac{1}{s} + \frac{s}{s^2 + 2}$$



Q Synthesize $Z(s)$ in Cauer-II form.

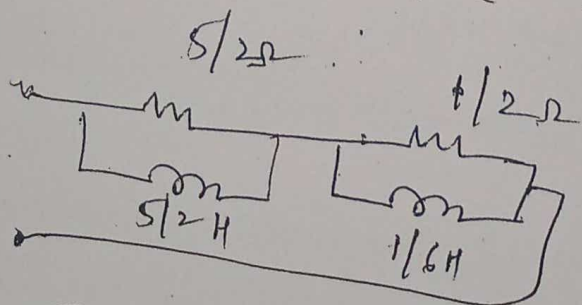
$$Z(s) = \frac{8s^3 + 10s}{s^4 + 6s^2 + 5}$$

2.

Q Show that ~~find~~ $F(s) = \frac{s(3s+8)}{(s+1)(s+3)}$ is R-L or
Realize in Foster-I form $\underline{\underline{R-L Imp.}}$

Solⁿ.

$$Z_{RL} = Y_{RC} = \frac{3s(s + \frac{8}{3})}{(s+1)(s+3)} = \frac{\frac{6}{2}s}{s+1} + \frac{\frac{8}{2}}{s+3}$$



Y_{RC} Z_{RL}

R-L Imp or R-C admittance.

Properties:

- ① P & Z lie on the -ve real axis.
- ② P & Z interlace each other along -ve real axis.
- ③ Residues of ^{poles of} $Z_{R-L}(s)$ or $Y_{R-C}(s)$ are real & -ve. however the residues of the poles of $\frac{Z_{RL}}{s}$ or $\frac{Y_{RC}}{s} \Rightarrow$ real & +ve.
- b) residues of the poles of $\frac{Y_{R-L}}{s}$ or Z_{R-C} must be real & +ve.
- ④ Singularity nearest to origin $\rightarrow$ zero.
- ⑤ Singularity nearest to $-\infty$ must be a pole.

Q $Z(s) \rightarrow \frac{s(s+2)(s+5)}{(s+1)(s+4)}$

Find R-L representation of Foster - I & II & Cauer - I & II.

Solⁿ

$$\frac{Z(s)}{s} = \frac{(s+2)(s+5)}{(s+1)(s+4)}$$

$$\begin{array}{r} (s^2 + 5s + 4) \overline{) s^2 + 7s + 10} \\ \underline{s^2 + 5s + 4} \\ 2s + 6 \end{array}$$

N/w Functions Responses

(4)

A) transform impedance at a port $\rightarrow$ ratio of voltage transform to current trans.

$$Z(s) = \frac{V(s)}{I(s)}$$

$$\text{transform admittance} = \frac{I(s)}{V(s)} = \frac{1}{Z(s)}$$

Imp. or adm. found at a given port is called a driving point imp. or adm.

$\rightarrow$ transform imp. or admittance of ports 1-1' & 2-2' are also called as I/P & O/P driving point-imp. or adm. resp.

$\rightarrow$ Comb of imp & admittance $\rightarrow$ immittance.

Elements	Imp.	adm.
R	R	$\frac{1}{R} = G$
L	sL	$\frac{1}{sL}$
C	$\frac{1}{sC}$	sC

$$\therefore Z(s) = R = sL = \frac{1}{sC}$$

$$Y(s) = \frac{1}{R} = \frac{1}{sL} = sC$$

(B) $\rightarrow$ x^{tr} funcⁿ is used to describe n/w which have at least 2 ports.

x^{tr} funcⁿ relates the transform of a quantity at one port to the transform of another quantity at another port.

Thus transfer func have the following possible forms:

① ratio of 1 voltage to another current or
1 current to another voltage.

② ratio of $\frac{Z(s) \text{ or } Y(s)}{I \text{ or } V}$ to another V .
or volt. transfer funcⁿ $G(s)$

③ ratio of 1 current to another I or
other I transfer funcⁿ $\alpha(s)$

following transfer funcⁿ for 2-port N/w.
are possible

(a) tr Imp funcⁿ $Z_{21}(s) = \frac{V_2(s)}{I_1(s)}$

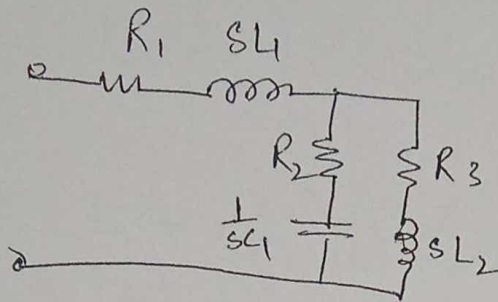
(b) tr adm funcⁿ $Y_{21}(s) = \frac{I_2(s)}{V_1(s)}$

(c) Volt. transfer funcⁿ $G_{21}(s) = \frac{V_2(s)}{V_1(s)}$

d) I tran. funcⁿ $\alpha_{21}(s) = \frac{I_2(s)}{I_1(s)}$

Inverse transfer funcⁿ $\rightarrow$ ratio of an i/p quantity to an o/p qty. (5)

Q1
(a)

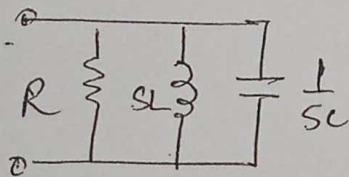


find Driving point
imp. funcⁿ or $Z(s)$.

$$Z(s) = (R_1 + sL_1) + \left[(R_2 + \frac{1}{sC_1}) \parallel (R_3 + sL_2) \right]$$

z

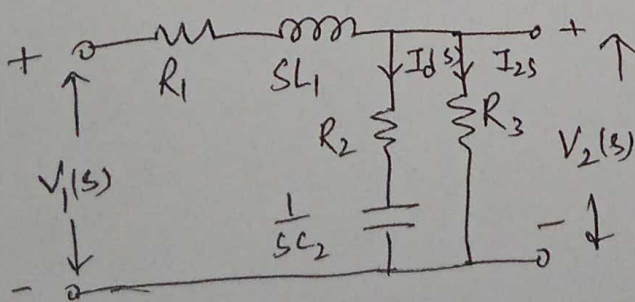
(b)



find Driving pt admittⁿ
or $Y(s)$.

$$Y(s) = \frac{1}{R} + \frac{1}{sL} + sC = \frac{sL + R + s^2 RLC}{sRL}$$

Ques For the ckt shown find n/w x^{fer} funcⁿ.



$$V_2(s) = I_2(s) \cdot R_3$$

$$= \left(R_2 + \frac{1}{sC_2} \right) I_0(s)$$

$$I_0(s) = \frac{V_2(s)}{R_2 + \frac{1}{sC_2}}$$

$$I_1(s) = I_2(s) + I_0(s)$$

$$V_1(s) = V_2(s) + (R_1 + sL_1) I_1(s)$$

$$Z_{21}(s) = \frac{V_2(s)}{I_1(s)}$$

$$I_1(s) = I_2(s) + I_0(s) = I_0(s) \left(\frac{I_2}{I_0} + 1 \right)$$

$$= I_0(s) \left(\frac{R_2}{R_3} + \frac{1}{R_3 s C_2} + 1 \right)$$

$$Z_{21}(s) = \frac{V_2(s)}{I_1(s)} = \frac{I_0(s) \left(R_2 + \frac{1}{s C_2} \right)}{I_0(s) \left(\frac{R_2}{R_3} + \frac{1}{R_3 s C_2} + 1 \right)}$$

$$Z_{21}(s) = \frac{R_3 (s C_2 R_2 + 1)}{s (R_2 + R_3) + 1}$$

$$(2) \quad Y_{21}(s) = \frac{I_2(s)}{V_1(s)}$$

$$V_1(s) = (R_1 + s L_1) I_1 + V_2(s)$$

$$= R_3 I_2(s) + (R_1 + s L_1) (I_2(s) + I_0(s))$$

$$= I_2(s) \left(R_3 + R_1 + s L_1 + (R_1 + s L_1) \left(\frac{R_3}{R_2 + 1/s C_2} \right) \right)$$

$$\frac{V_1(s)}{I_2(s)} = \frac{s C_2 R_2 + 1}{L_1 C_2 (R_2 + R_3) s^2 + [C_2 (R_1 R_2 + R_2 R_3 + R_3 R_1) + L_1] s + R_1 + R_3}$$

(6)

$$G_{21}(s) = \frac{V_2(s)}{V_1(s)} = \frac{V_2(s)}{I_1(s)} \cdot \frac{I_2(s)}{V_1(s)} \cdot \frac{I_1(s)}{I_2(s)}$$

$$= Z_{21}(s) \cdot Y_{21}(s) \cdot \frac{I_1(s)}{I_2(s)}$$

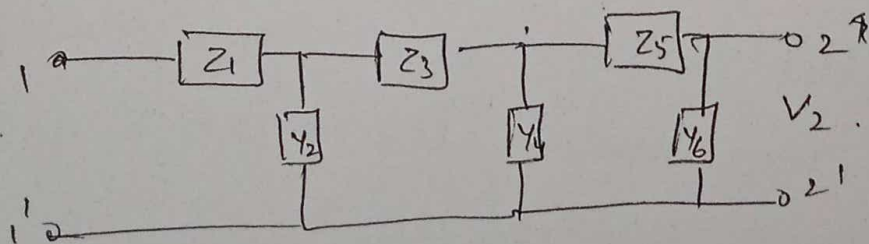
$$\begin{aligned} I_1(s) &= I_2(s) + I_0(s) = I_2(s) \left(1 + \frac{I_0}{I_2} \right) \\ &= I_2(s) \left(1 + \frac{R_3}{R_2 + \frac{1}{sC_2}} \right) \end{aligned}$$

$$\frac{I_1}{I_2} = \frac{(sR_2C_2 + 1) + sC_2R_3}{sR_2C_2 + 1}$$

$$G_{21} =$$

$$\& \alpha_{21} = \frac{I_2}{I_1} = \frac{sR_2C_2 + 1}{(sR_2C_2 + 1) + sC_2R_3}$$

(c) Calc. of N/w funcⁿ for ladder N/w:



$$Z = Z_1 + \frac{1}{Y_2 + \frac{1}{Z_3 + \frac{1}{Y_4 + \frac{1}{Z_5 + \frac{1}{Y_6 + \dots}}}}}$$

Poles & Zeros of N/w funcⁿ:

$$T(s) = \frac{N(s)}{D(s)} = \frac{a_n s^n + a_{n-1} s^{n-1} + \dots + a_0}{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}$$

(N/w funcⁿ)

~~where~~ $N(s) = 0$ has n roots $\Rightarrow$ zeros of N/w funcⁿ $T(s)$.

$D(s) = 0$ has m roots $\Rightarrow$ poles of N/w funcⁿ $T(s)$.

$$T(s) = K \frac{(s-z_1)(s-z_2)\dots(s-z_n)}{(s-p_1)(s-p_2)\dots(s-p_m)}$$

~~Zeros~~

$\rightarrow$ value of p & z of $T(s)$ & hence their location in s -plane completely specify the n/w funcⁿ except scalar factor K .

$p \rightarrow (X)$ in s -plane.

$z \rightarrow (O)$

Case I $\rightarrow r$ poles ($r < m$) or r zeros ($r < n$).

have same value, the p or z is said to be of multiplicity r .

Case II $\rightarrow n > m$, we have poles at ∞ of multiplicity or degree of $n-m$.

Case III

$n < m$ we have zeros at ∞ of (7)
degree $(m-n)$.

for any rational n/w funⁿ.

no. of finite p + no. of p at ∞

= no. of finite z + no. of z at ∞ .

So tot. no. of p = tot. ^{no.} of z .

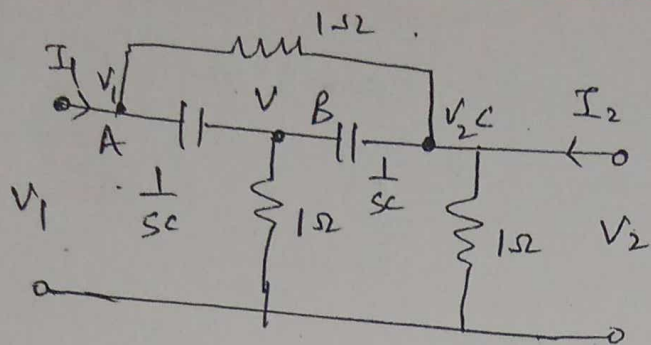
Q Obtain pole-zero location.

$$T(s) = \frac{(2s+4)(s+4)}{s(s+1)(s+3)}.$$

$$P \rightarrow s = 0, s = -1, s = -3$$

$$Z \rightarrow s = -2, s = -4, s = \infty.$$

Q



Find driving pt
imp. $Z_{11}(s)$; tr. imp.
funⁿ & volt. tr. funⁿ

Solⁿ

Apply KCL at A.

$$I_1 = \frac{V_1 - V}{1/s} + \frac{V_1 - V_2}{1}$$

$$I_1 = (s+1)V_1 - sV - V_2 \quad \text{--- (1)}$$

At B

$$\frac{V_1 - V}{1/s} = \frac{V}{1} + \frac{V - V_2}{1/s}$$

$$sV_1 - (2s+1)V + sV_2 = 0 \quad \text{--- (2)}$$

$$\text{At C} \quad \frac{V_1 - V_2}{1} + \frac{V - V_2}{1/s} = \frac{V_2}{1}$$

$$V_1 + sV - (s+2)V_2 = 0 \quad \text{--- (3)}$$

$$Z_{21} = \frac{V_2}{I_1} = \frac{s^2c^2 + 2sc + 1}{2s^2c^2 + 5sc + 1}$$

$$Z_{11} = \frac{V_1}{I_1} = \frac{s^2c^2 + 5sc + 2}{2s^2c^2 + 5sc + 1}$$

$$G_{21} = \frac{V_2}{V_1} = \frac{s^2c^2 + 2sc + 1}{s^2c^2 + 5sc + 2}$$